

LAWRENCE TECHNOLOGICAL UNIVERSITY · COLLEGE OF ARCHITECTURE &
DESIGN

The *Invisible* Architecture & the Listening Room.

Acoustics, daylight, and electrical illumination – the three forces that shape a room without ever being drawn on a plan. Ten weeks, fifteen students, one five-hundred-seat auditorium designed through the ear, the eye, and the wire.

TERM

Summer 2026

05.20 → 07.22 · 10 weeks · online

INSTRUCTOR

Kurt Neiswender, AIA

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MEETS

Wed · 19:30–23:35

Zoom – cameras on, 100%

COHORT

15 students

Graduate · lecture & lab

§ 01 The *Invisible* Architecture.

Walls can be photographed. Sound and light cannot. Yet the experience of a room – whether a singer’s breath reaches the back row, whether a lobby feels like arrival or like waiting – is governed almost entirely by these invisible forces. They are present in every section drawing without ever appearing in one. This course treats them as the primary clients of the building.

For ten weeks, every student will design the same building: a *500-seat auditorium with a prefunction room*. Each student receives the same starting kit in Week 1 – generic seating diagrams (shoebox, fan, vineyard, horseshoe), a stage orientation, a site, a climate, and a program. From that kit, you will shape the hall three times over – first through the ear, then through the eye, then through the wire – with a fourth week reserved for tightening, and a tenth for the curtain.

Acoustics comes first because the shape of the room is acoustically determined: once the section and plan are tuned for sound, the daylight and lighting strategies follow. Daylight comes second, compressed to two weeks – its deeper treatment lives in the HVAC and Water Systems course. Lighting and electrical close the loop: the design we hear and see is finally *wired, dimmed, and powered*. Week 9 is yours for integration; Week 10 is the encore.

I.

ACT ONE • WEEKS 1–3

Acoustics & *Room Shape*

Sound first. Section, plan curvature, ceiling slope, seat rake – shaped with the ear as the primary client. Sabine, RT60, early reflections, NC targets, STC separation from back-of-house.

Wks 01 • 02 • 03 // MEEB ch.7 • 22
• 23

III.

ACT THREE • WEEKS 6–8

Lighting & *Electrical*

House lighting, theatrical positions, dimming & controls, panel schedule and circuit groupings, PV offset. All deliverables are *modeled drawings* in Rhino or Revit – no hand-drawn circuits.

Wks 06 • 07 • 08 // MEEB ch.15•1•2
• 16 • 17 • 25 • 26 • 28 • 29

II.

ACT TWO • WEEKS 4–5

Daylight & the *Prefunction*

With the shell stable, we open it – selectively. Prefunction as a daylit room of arrival; the hall studied for clerestory or controllable apertures that don't compromise blackout or NC.

Wks 04 • 05 // MEEB ch.10 • 6

Finale

TIGHTENING & ENCORE • 9–10

Integration & *Curtain*

Week 9 is a working week – rehearse, refine, integrate across systems, desk crits with the instructor. Week 10 is the final presentation. A revision window follows before final grading submission.

Wks 09 • 10 // assemble & present

The *Decision Log*.

Before the schedule, a word on the artifact that runs underneath it. The Decision Log is the primary learning record of the course – the place where your *thinking* about the hall is captured as it develops. It is not a transcript of what you did; it is the reasoning behind why you did it, and what you traded away.

You will maintain the log continuously across all ten weeks. It is *not* a separate deliverable each week – weekly submissions focus on the modeled work – but the log is the evidence base reviewed at the close of each act and at the encore. Roughly 2–4 short entries per week is typical; quality of reasoning matters more than count.

WEEK / ACT

Week 02 · Act I – Acoustics

SYSTEM

Acoustics – Room Shape

SPACE

Main Hall

DECISION

Adopted a 14° ceiling splay over the rear two-thirds of the hall and lowered the proscenium reflector to 28 ft.

REASONING

Pachyderm showed late lateral energy was weak in row 20+; the splay redirects first reflections to the rear seating. Hand-calc Sabine RT60 at 1 kHz moved from 1.9 s to 1.6 s – closer to the speech/music compromise target band of 1.4–1.7 s.

TRADE-OFF

Lost 6 ft of fly height above the proscenium; will reduce theatrical lighting throw options – revisit in Act III.

WHAT I'D
RECONSIDER

(Filled in at Act I Review.) Should the splay be steeper at the back wall to avoid focusing echoes? Test in Pachyderm before Act II.

A template document is provided in the Week 1 Canvas module. The log is evaluated qualitatively – the criterion is *quality of reasoning*, not whether your first decision was correct. A student who identifies a flawed decision, explains why, and corrects it has demonstrated more learning than one who happened to be right by accident.

Ten-Week *Roadmap.*

Each week opens with two micro-lectures – roughly 30 min on the assigned book content and 30 min on supporting material (precedents or software tutorials). The long middle of every session is *lab*, with breakout groups and a circulating instructor. Each row below tells you exactly what to read, what to expect in lecture, what the lab breakout will look like, and what is due.

01

WED 05 • 20

19:30–23:35

ACT I • ACOUSTICS

Sound, Hearing, and the *Listening Room.*

Course orientation; the 500-seat program & starting plan kit; fundamentals of sound – frequency, dB, equal-loudness, NC criteria.

— READ • BEFORE

MEEB *Ch. 7 – Sound, Hearing, & Acoustical Comfort*. Complete the intake survey on Canvas.

— MICRO-LECTURES

Book (30 min): sound fundamentals, NC criteria.

Support (30 min): precedent tour – Boston Symphony, Berlin Philharmonie, Bayreuth, Disney Hall.

— LAB BREAKOUT • EXAMPLE

Groups of 3–4: each group is assigned one of the four seating geometries (shoebox / fan / vineyard / horseshoe). Discuss acoustic implications, sketch sectional response. Each *student* then commits individually to a starting scheme and begins their Rhino model.

— DELIVERABLE • DUE TUE 05 • 26

Submit to Canvas:

- Annotated plan + section (2 schemes)
- Short written rationale for chosen seating geometry

02

WED 05•27

19:30-23:35

ACT I • ACOUSTICS

Room Shape, RT60 & *Sabine*.

Absorption coefficients; NRC tables; targeting RT60 for speech vs. music; ceiling clouds, splays, the geometry of early reflections.

— READ • BEFORE

MEEB *Ch. 22 – Sound in Enclosed Spaces.*

— MICRO-LECTURES

Book: Sabine's equation, NRC, absorption math.

Support: Pachyderm setup tutorial – from Rhino surface to first ray-trace.

— LAB BREAKOUT • EXAMPLE

Pairs: partner with one other student. Run each other's Rhino models through Pachyderm, compare RT60 at 500 / 1k / 2k Hz against target band. Cross-critique the geometry, then hand-calc RT60 to verify the simulation.

— DELIVERABLE • DUE TUE 06•02

Submit:

- RT60 worksheet (hand + Pachyderm)
- Annotated ray-trace screenshots
- Updated Rhino model

Noise Control & *Act I Pin-Up*.

STC, mass law, flanking paths; separating loading dock and mechanical from the hall; door seals, vestibules. Closes with Act I review.

— READ • BEFORE

MEEB *Ch. 23 – Building Noise Control*.

— MICRO-LECTURES

Book: STC, mass law, isolation strategies.

Support: pin-up format and reviewer rubric.

— LAB BREAKOUT • EXAMPLE

First half (60 min): STC calculation for the back-of-house party wall; assembly spec-up.

Second half (60 min): *Act I Review* – each student presents room shape + RT60 + STC strategy in 5 min; cohort + instructor critique.

— DELIVERABLE • ACT I PACKAGE

Submit:

- Acoustic shape (plan + section) with material tags
- RT60 band & STC wall assembly
- Act I review slides & presentation

Solar Geometry & the *Aperture Question.*

Sun path, altitude/azimuth, site latitude; where (and whether) to put glass on a performance hall; prefunction as a separate daylight problem.

— READ • BEFORE

MEEB *Ch. 10 – Daylighting* (pp. 1–25).

— MICRO-LECTURES

Book: solar geometry, daylight metrics (sDA, ASE, DF).

Support: LightStanza walkthrough – importing Rhino models into the web app; Ladybug sun-path tutorial.

— LAB BREAKOUT • EXAMPLE

Groups of 3: each student brings their acoustic Rhino mass into LightStanza or Ladybug. Run solstice + equinox sun studies; tag daylight zones on the prefunction; share findings; vote on the most promising aperture stance for the hall.

— DELIVERABLE • DUE TUE 06 • 16

Submit:

- Sun-study screenshots (solstice + equinox)
- Daylight-zone diagram (prefunction + hall)
- Aperture-stance memo (1 page)

05

WED 06 • 17

19:30–23:35

ACT II • DAYLIGHT

ACT II REVIEW

Visual Comfort, Glare & *Act II Pin-Up*.

Human visual adaptation; the 10:1 contrast rule; clerestory vs. sidelight vs. baffled aperture; reconciling daylight with hall blackout. Closes with Act II review.

— READ • BEFORE

MEEB *Ch. 6 – Light & Visual Comfort*;
LightStanza help docs.

— MICRO-LECTURES

Book: visual adaptation, glare, sDA/ASE thresholds.

Support: precedents – Kimbell Art Museum, Glyndebourne lobby, Aspen Music Tent.

— LAB BREAKOUT • EXAMPLE

Individual (60 min): parametric WWR sweep in LightStanza for prefunction; controlled aperture study for hall.

Second half (60 min): *Act II Review* – 5-min pin-ups; cohort + instructor critique focused on glare, blackout integrity, and aperture reasoning.

— DELIVERABLE • ACT II PACKAGE

Submit:

- LightStanza comparison sheets (min. 2 scenarios)
- Annotated daylight plan + reflectance specs
- Act II review slides & presentation

Lamps, Luminaires & *Theatrical Positions.*

LED, CCT, CRI, efficacy; house light vs. work light vs. performance rig; catwalks, FOH positions, box-boom, ellipsoidal vs. wash; IES files.

— READ • BEFORE

MEEB *Ch. 15 Pt. 1 & 2 – Lamps, Luminaires, Controls.*

— MICRO-LECTURES

Book: LED / CCT / CRI / efficacy; lamp families.

Support: a real venue rig walk-through – FOH catwalk plot from a working road house.

— LAB BREAKOUT • EXAMPLE

Groups of 3, each takes one family: (a) house downlights, (b) prefunction decorative + general, (c) FOH stage rig. Each group sources 2–3 real luminaires (IES + spec sheet), then re-pairs to share findings so every student leaves with all three families.

— DELIVERABLE • DUE TUE 06•30

Submit:

- Spec packet (3 luminaire families, IES included)
- Selection rationale – CCT, CRI, throw, why

07

WED 07•01

19:30-23:35

ACT III • LIGHTING/ELEC

Lighting Design & *Controls.*

Lumen method & zonal cavity; uniformity; theatrical dimming (DMX overview); occupancy & daylight controls in prefunction; emergency & egress lighting.

— READ • BEFORE

MEEB *Ch. 16 & Ch. 17 – Lighting Design & Applications.*

— MICRO-LECTURES

Book: lumen method & zonal cavity worked example.

Support: tutorial – LightStanza electric workflow
OR Revit luminaire placement + IES (students pick their tool).

— LAB BREAKOUT • EXAMPLE

Pair-work by tool: Rhino+LightStanza pairs and Revit pairs cluster separately. Place house + prefunction luminaires, generate illuminance map, iterate against IES targets (30–50 fc reading task, lower for hall). Cross-tool comparison at the debrief.

— DELIVERABLE • DUE TUE 07•07

Submit:

- Illuminance maps (house + prefunction)
- Controls strategy annotated plan
- Updated luminaire schedule

08

WED 07•08

19:30–23:35

ACT III • LIGHTING/ELEC

ACT III REVIEW

Electrical, Panels, PV & *Act III Pin-Up.*

V/I/R/P, AC/DC; load calc & LPD; the path utility → panel → branch; rig power as a special load; PV array sized to offset the lighting load.

— READ • BEFORE

MEEB *Ch. 25, 26, 28; Ch. 29 (Photovoltaics).*

— MICRO-LECTURES

Book: V/I/R/P, load calc, LPD compliance, PV basics.

Support: real panel schedule walk-through from a small commercial venue.

— LAB BREAKOUT • EXAMPLE

First half (60 min): produce the *lighting + power plan as a modeled drawing* in Rhino or Revit – luminaire layout, circuit groupings annotated, panel locations, load summary table, PV sizing.

Second half (60 min): *Act III Review* pin-up.

— DELIVERABLE • ACT III PACKAGE

Submit:

- Lighting plan (Rhino or Revit)
- Power plan + panel schedule
- Load summary + PV calc
- Act III review slides & presentation

09

WED 07 • 15

19:30–23:35

INTEGRATION • TIGHTENING

Cross-System *Tightening*.

A working week. Refine, integrate, rehearse. Resolve cross-system trade-offs: where did acoustics constrain daylight? Where did the rig conflict with the ceiling cloud?

— READ • BEFORE

Review all course materials; assemble book draft.

— MICRO-LECTURES

Book recap (30 min): cross-system integration case study – a real auditorium where acoustics, daylight, and lighting interlock.

Support (30 min): final book + presentation template walk-through.

— LAB BREAKOUT • EXAMPLE

Open studio: students work independently on the integrated book and presentation. Instructor holds 10-minute 1:1 desk crits in rotating breakout rooms. Optional dry-run slot in the final 30 minutes.

— DELIVERABLE • DUE TUE 07 • 21

Submit:

- Draft final book (all three acts assembled)
- Presentation outline / slide draft
- Cross-system trade-off memo

10

WED 07•22

19:30-23:35

THE ENCORE • FINAL PRESENTATIONS

The Hall, *Heard, Seen, Powered.*

Final presentations. Each student presents the auditorium across all three acts. 10 min present + 5 min Q&A. Guest critic invited.

— READ • BEFORE

Rehearse. Finalize slide deck and walkthrough video.

— FORMAT

15 presentations × 15 min = full session. Order announced Week 9. Guest critic joins for the full evening.

— LAB BREAKOUT • EXAMPLE

No breakouts this week. The session *is* the encore. Cohort remains together for all presentations – peer engagement counts toward participation.

— FINAL BOOK • DUE SUN 07•26 • 11:59 PM

A short revision window follows the encore – grades are due to the registrar 07•29. Incorporate critique feedback, then submit.

Submit:

- Full project book (all three acts + integration)
- Three system summary sheets
- Annotated plan set (acoustic / daylight / lighting+power)
- PV calculation
- 5-min walkthrough video
- Complete Decision Log

The Session — *Lab-Forward.*

Every Wednesday session is built around two micro-lectures and a long working lab. The micro-lectures keep theory tight so the bulk of the four hours can be spent making, testing, and critiquing the building you are designing.

19:30 · 19:40	10 min	Arrival & week framing – why this week matters for your hall right now.
19:40 · 20:10	30 min	<i>Micro-Lecture A – Book Content.</i> The week's MEEB chapter, distilled, with a worked example.
20:10 · 20:20	10 min	Break – cameras off, move around.
20:20 · 20:50	30 min	<i>Micro-Lecture B – Support.</i> Precedent, software tutorial, or guest demo.
20:50 · 21:00	10 min	Lab framing – instructor sets up the lab tasks and breakout assignments.
21:00 · 23:20	140 min	<i>Lab.</i> Breakout rooms of 3–4. Instructor circulates. See each week for the breakout example.
23:20 · 23:35	15 min	Whole-group debrief – one share per room; instructor sets next-week ask.

Breakout composition is consistent for Acts I and II, then reshuffled at Week 6 for the technical integration phase. All sessions are recorded and posted within 24 hours; recordings are chaptered (Lecture A, Lecture B, Lab Intro) for review. Watching a recording does *not* substitute for live presence – the lab is the course.

Primary — *Rhino*

Rhino + Grasshopper is the connective tissue of the semester. The acoustic shape, the daylight mass, and the lighting plan can all live in one model. Pachyderm runs inside Rhino; Ladybug runs in Grasshopper; LightStanza ingests the Rhino model directly via its web app.

Required Weeks 2–9.

Daylight + Lighting — *LightStanza*

LightStanza handles both daylight and artificial-lighting analysis in a single environment. Two access paths: **(a)** upload your Rhino model to the LightStanza web app, or **(b)** use the Revit plug-in if you prefer the BIM workflow.

Account setup in Week 4. Notify instructor of any license issues immediately.

Optional — *Revit*

Revit is fully supported for students who prefer it. Luminaire placement, IES analysis, panel schedules, and the Week 8 lighting+power plan can all be produced in Revit. Use *Rhino.Inside.Revit* to move acoustic geometry across.

Academic license via Autodesk Education.

Supporting tools: Pachyderm (acoustic), Ladybug (solar), Excel or Sheets (load summaries, PV calc). Software access should never block a deliverable – email the instructor immediately if you hit a license wall.

Evaluation & *Grading.*

Three Act Reviews and an Encore. Each Act Review evaluates the package due at the close of that act — the modeled drawings, the analysis, and the Decision Log section. The Encore is the final presentation plus the assembled book due one week after, during the revision window.

COMPONENT	WEIGHT
Weekly Deliverables – Weeks 1, 2, 4, 6, 7, 9 (6 × 5%)	30%
Act I Review – Acoustics (Week 3)	10%
Act II Review – Daylight (Week 5)	10%
Act III Review – Lighting/Electrical (Week 8)	10%
Encore – Final Presentation & Book (Week 10 + revision week)	30%
Participation, Cameras, Peer Engagement	10%
Total	100%

Weekly Deliverables are evaluated on a 5-point scale (completeness + reasoning quality). Act Reviews are graded against a rubric posted in Canvas covering: technical content, modeling quality, Decision Log reasoning, and presentation clarity. The Encore is graded on integrated thinking across all three acts — the question is whether the hall *holds together* as a designed object.

GRADING SCALE — GRADUATE

A 94–100	A - 90–<94	B + 87–<90
B 84–<87	B - 80–<84	C + 77–<80
C 74–<77	C - 70–<74	F <70

C- is the lowest passing graduate mark per CoAD policy. Anything below 70 is recorded as F and the course must be retaken to satisfy degree requirements.

GRADING SCALE — UNDERGRADUATE

A 94–100	A - 90–<94	B + 87–<90	B 84–<87
B - 80–<84	C + 77–<80	C 74–<77	C - 70–<74
D + 67–<70	D 64–<67	D - 61–<64	F <61

This course is co-listed with undergraduate enrollment. Undergraduate students follow the standard LTU letter scale above; F is failing.

The instructor reserves the right to adjust the low-end average cut-off score based on the statistical distribution of final semester averages. Late work accepted with a 48-hour grace period at no penalty; beyond, a 20%-per-week deduction applies. Work more than two weeks late will not be accepted without prior approval.

§ 08 Canvas — *Online Course Management.*

The course syllabus, calendar, assignments, additional readings, and notes are posted on Canvas. Students are responsible for monitoring the site regularly. You are expected to have your camera on 100% of the class session to receive attendance and participation credit.

Each week's Canvas module follows an identical structure so you always know where to look:

- **0. Before Class** – required reading and any short prep video.
- **1. This Week's Concepts** – lecture slides and worked examples (posted after class).
- **2. Lab Tasks** – step-by-step task list for the in-class lab (posted before class).
- **3. Deliverable** – what to submit, file format, due date (Tuesdays 11:59 PM), submission link.
- **4. Discussion Board** – optional, ungraded; instructor posts a weekly prompt.

A **Weekly Snapshot Thread** runs the full 10 weeks. Each student posts one image per week – a screenshot of their current project work. This is required but ungraded. It creates a visible record of your hall's evolution and helps the instructor identify questions early.

§ 09 Attendance, *Cameras* & Conduct.

Attendance & the *20% rule*

LTU policy permits a maximum of 20% absence. For a once-weekly 10-week course, that is **2 absences** before failure.

Warning sequence:

- 1st absence – instructor email.
- 2nd absence – written warning + required check-in + registrar notification.
- 3rd absence – failing grade for non-attendance.

Documented emergencies handled case-by-case; communicate before, not after. Watching the recording does not substitute for live presence in this lab-driven format.

Camera Policy — *on, 100%*

This is a small, lab-driven cohort. Cameras are **on for the full session** – lecture, demo, breakout, debrief – to receive attendance and participation credit. The only exception is the mid-session break.

If a camera-related circumstance arises (bandwidth, family obligation, illness), email the instructor before class. A blank tile mid-session is read as an absence.

The lab is collaborative space-shaping. We need to see your face, your hands, and your work.

THE STUDENT COMPANION – CONDUCT, HONESTY, DEPARTMENT POLICIES

All college policies and procedures have been compiled into a single document, *The Student Companion*. All students are responsible for reading and understanding these policies. The Companion is posted online:

https://www.ltu.edu/architecture_and_design/student_resources.asp

Topics covered: the Studio Code, plagiarism, attendance and late work, participation.

GRADUATE ONLINE COURSES – ATTENDANCE, PARTICIPATION, ENGAGEMENT

CoAD graduate courses are comprised of asynchronous and synchronous educational engagement structures so that knowledge can be explored in a variety of ways. To maintain the high, rigorous standards of the CoAD academic environment, consistent attendance and participation in both modalities is a necessity. Required synchronous class sessions will be held within the class times listed on Banner; other meetings may be arranged by student-faculty agreement. In the case of conflicts between a listed course time, synchronous content delivery, and extra-curricular student responsibilities, the student should communicate directly with the faculty member. The student is responsible to compensate for these scheduling conflicts during the semester. Since graduate work is assessed on performance, a lack of engagement through absence will inevitably impact educational outcomes.

CATALOG DESCRIPTION

The course reviews the theories of sound perception, analysis of architectural acoustics, room acoustics, sound isolation, sound absorption, electronic sound reinforcement, and the study of acoustic properties of materials and room shapes to provide speech privacy, enhance speech clarity, enrich music spaces and control noise. The course also addresses basic electrical theory, utility connection options, emergency backup, distribution, and safety considerations as they relate to building construction. Electrical lighting is studied as part of an assimilated section covering the fundamentals of illumination, integrating natural daylighting and electrical lighting design for energy conservation. Class projects include investigations into course subjects to further enhance understanding of real-work applications.

PREREQUISITES

Architecture Foundations Studio 2 – ARC 5024

Architecture Interiors Studio 1 – ARC 5113

NAAB PROGRAM CRITERIA

Environmental Systems (SC.4B) – principles of environmental system design; application to problems involving solar geometry, daylight, natural ventilation, heating and cooling, indoor air quality, lighting systems, and acoustics; and analysis across different geographic regions.

Building Service Systems (SC.4E) – principles of building service systems (mechanical, plumbing, fire protection, electrical, lighting); appropriate application; performance analysis through quantitative reasoning; application to problems using advanced technologies.

COURSE OBJECTIVES

- **Environmental Systems:** demonstrate principles of environmental systems design, how design criteria vary by geographic region, and the tools used for performance assessment – including daylighting, lighting systems, and acoustics. (SC.4B)
- **Building Service Systems:** understand the basic principles and appropriate application and performance of building service systems, including lighting and electrical. (SC.4E)

ADDITIONAL LEARNING OUTCOMES

- **Ecological Knowledge** – dynamics between built and natural environments; resilience.
- **Health, Safety & Welfare** – impact of the built environment on human comfort and safety at multiple scales.
- **Building Integration** – design decisions that integrate envelope, structural, environmental control, and life-safety systems with measurable performance outcomes.

§ 11 Texts & *References.*

Required

Mechanical & Electrical Equipment for Buildings (MEEB), 13th ed.
Grondzik & Kwok · Wiley, 2019.

Chapter scans for assigned weeks are posted on Canvas; physical copy strongly recommended.

Recommended

- Livingston – *Designing with Light*.
- Long – *Architectural Acoustics* (selected chapters, Canvas PDF).
- La Roche – *Carbon-neutral Architectural Design*, 2e.
- Beranek – *Concert Halls and Opera Houses* (Act I precedents).
- The Coffee Sketch Podcast – coffeesketchpodcast.com

A n auditorium is one of the few building types whose quality is measurable at every register of the senses. Sound and light are *invisible* — they leave no line on a plan — yet they decide whether the room works. Over ten weeks we move through the hall in the order it should be designed: heard first, seen second, operated last.

Calculations are not the goal. The hall is. Every reverberation time, every footcandle, every wired circuit exists to answer a question about the room you are making. Bring the questions to the lab. The calculations will follow.